JAVA_PRACTICE_TEST_2

Total points 36/87

The respondent's email (**sumitnarwade98@gmail.com**) was recorded on submission of this form.

Consider the following Java code snippet. What will be the output when it is executed? *1/1

```
class Person {
    String name;

    Person(String name) {
        this.name = name;
    }
}

public class Test {
    public static void main(String[] args) {
        Person p1 = new Person("Alice");
        Person p2 = p1;
        p2.name = "Bob";
        System.out.println(p1.name);
    }
}
```

- A) Alice
- B) Bob
- C) Compilation Error
- D) null

In Java, where are primitive types and objects stored in memory? * 1/1

- A) Primitive types are stored in heap memory, while objects are stored in stack memory.
- B) Both primitive types and objects are stored in the stack memory.
- C) Primitive types are stored in stack memory, while objects are stored in heap memory.
- D) Both primitive types and objects are stored in the heap memory.

What is the main difference between the Thread class and the Runnable interface? * 1/1

- a) Thread is used to create threads, while Runnable is used to define the task for the thread.
- b) Runnable is a class that provides a mechanism for thread management, while Thread is an interface.
- c) Thread can be implemented by a class, while Runnable can only be extended by a class.
- d) Thread is used for non-blocking threads, while Runnable is used for blocking threads.

Which of the following methods is used to make a thread wait until another thread sends a signal to continue its execution? * 1/1

- a) wait()
- b) notify()
- c) sleep()
- d) join()

Which of the following is a characteristic feature of Hashtable? * 1/1

- [x] a) Hashtable is synchronized and thread-safe, but does not allow null values. ✓
- [ ] b) Hashtable is thread-unsafe and allows duplicate keys.
- [ ] c) Hashtable allows null keys and values.
- [ ] d) Hashtable does not allow duplicate values, but allows duplicate keys

In the Producer-Consumer problem, which of the following statements is true regarding the role of wait(), notify(), and notifyAll()? * 0/1

- [x] a) wait() is used to pause the producer when the buffer is full, and notify() is used to notify the consumer. ✗
- [ ] b) notify() is used to pause the consumer when the buffer is empty, and notifyAll() is used to notify all producers.
- [ ] c) notifyAll() is used when a producer adds an item, and wait() is used to pause the consumer when there are no items.
- [ ] d) notify() is used to pause the producer when the buffer is full, and wait() is used to notify the consumer.

Correct answer

- [x] c) notifyAll() is used when a producer adds an item, and wait() is used to pause the consumer when there are no items.

Given the following code, what concept is demonstrated by the Vehicle class, which contains the speed() method, and the Car class that overrides it? *0/1

```
class Vehicle {

   void speed() {

      System.out.println("Vehicle is moving");

   }

}


class Car extends Vehicle {

   @Override

   void speed() {

      System.out.println("Car is moving faster");

   }

}


public class Test {

   public static void main(String[] args) {

      Vehicle v = new Car();

      v.speed();

   }

}
```

- A) Encapsulation
- B) Inheritance ✗
- C) Polymorphism
- D) Abstraction

Correct answer

- C) Polymorphism

PRN *

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✗ Which of the following best describes the difference between shallow copy and deep copy in Java? *0/1

- A) A shallow copy creates a new object with the same reference as the original, while a deep copy creates a completely independent copy of the original object and its fields.
- B) A shallow copy copies only primitive data types, while a deep copy copies objects only.
- C) A shallow copy copies fields by reference, while a deep copy only copies immutable fields.
- D) A shallow copy is used for cloning objects, while a deep copy is used to serialize objects. ✗

Correct answer

- A) A shallow copy creates a new object with the same reference as the original, while a deep copy creates a completely independent copy of the original object and its fields.

What is the result of the following code, and why? class A { public void method() throws Exception { throw new Exception("Exception from A"); }} class B extends A { public void method() throws RuntimeException { throw new RuntimeException("Exception from B"); }} public class Test { public static void main(String[] args) { A obj = new B(); try { obj.method(); } catch (Exception e) { System.out.println("Caught Exception: " + e.getMessage()); } }} *1/1

A) Caught Exception: Exception from A
B) Caught Exception: Exception from B
C) Compilation error
D) Runtime exception not caught45. What is the result of the following code, and why? class A { public void method() throws Exception { throw new Exception("Exception from A"); }} class B extends A { public void method() throws RuntimeException { throw new RuntimeException("Exception from B"); }} public class Test { public static void main(String[] args) { A obj = new B(); try { obj.method(); } catch (Exception e) { System.out.println("Caught Exception: " + e.getMessage()); } }}

- (x) A) Caught Exception: Exception from A ✓
- ( ) B) Caught Exception: Exception from B
- ( ) C) Compilation error
- ( ) D) Runtime exception not caught

What happens when Thread.yield() is called? * 0/1

- a) It terminates the current thread.
- b) It pauses the current thread and allows other threads of the same priority to run.
- c) It suspends the current thread indefinitely.
- d) It causes the thread to sleep for a specified time

Correct answer

- b) It pauses the current thread and allows other threads of the same priority to run.

What is the primary advantage of using Generics in Java? * 1/1

- a) It allows dynamic binding of types at runtime.
- b) It enables the use of primitive types in collections.
- c) It helps ensure type safety by providing compile-time type checking.
- d) It eliminates the need for reflection in Java programs.

What is the primary difference between StringBuilder and StringBuffer in Java? *0/1

- A) StringBuilder is synchronized, while StringBuffer is not.
- B) StringBuffer is synchronized, while StringBuilder is not.
- C) StringBuilder and StringBuffer are identical in functionality and performance.
- D) StringBuilder can only be used for string concatenation, while StringBuffer can be used for general-purpose string manipulation.

Correct answer

- B) StringBuffer is synchronized, while StringBuilder is not.

What will be the output of the following code involving StringBuffer? StringBuffer sb = new StringBuffer("Java");sb.append(" Programming");String str = sb.toString();System.out.println(str); *0/1

- A) Java
- B) Java Programming
- C) Programming Java
- D) Compilation error

Correct answer

- B) Java Programming

When compiling a Java program, which of the following best describes the relationship between the Java source code and the Java bytecode? *0/1

- A) Java source code is compiled into machine code by the Java compiler and executed directly by the OS.
- B) The Java compiler translates the Java source code into Java bytecode, which is then executed by the JVM.
- C) Java source code is compiled into an intermediate code that is executed directly by the CPU.
- D) Java source code is compiled into an executable, which does not require the JVM to run.

Correct answer

- B) The Java compiler translates the Java source code into Java bytecode, which is then executed by the JVM.

. Consider the following code: import java.util.*; public class Main { public static void main(String[] args) { Map<Integer, String> map = new LinkedHashMap<>(); map.put(3, "Three"); map.put(1, "One"); map.put(2, "Two"); map.put(4, "Four"); System.out.println(map); }} *1/1

What will be the output?

- a) {1=One, 2=Two, 3=Three, 4=Four}
- b) {3=Three, 1=One, 2=Two, 4=Four}
- c) {1=One, 3=Three, 2=Two, 4=Four}
- d) {4=Four, 3=Three, 2=Two, 1=One}

What happens when the finalize() method is called on an object in Java? * 0/1

- A) It is automatically called by the garbage collector before the object is reclaimed.
- B) It must be manually invoked by the developer to free up resources.
- C) It can only be called once, and it cannot be overridden in any subclass.
- D) It is called only when the object is explicitly dereferenced by the developer.

Correct answer

- A) It is automatically called by the garbage collector before the object is reclaimed.

Which of the following is true about the Java Virtual Machine (JVM) and its platform independence? * 0/1

- A) The JVM is platform-independent; Java bytecode is compiled to machine code specific to each platform.
- B) The JVM must be re-written for each platform, but once written, it allows Java bytecode to be executed on all systems.
- C) Java programs are compiled directly into platform-specific binaries.
- D) Java bytecode is platform-specific, and JVM makes the code independent of the hardware architecture.

Correct answer

- B) The JVM must be re-written for each platform, but once written, it allows Java bytecode to be executed on all systems.

What is the purpose of the throw keyword in Java? * 0/1

- A) It is used to propagate exceptions up the call stack.
- B) It is used to explicitly throw exceptions within a method.
- C) It is used to declare that a method may throw an exception.
- D) It is used to handle an exception after it has been caught.

Correct answer

- B) It is used to explicitly throw exceptions within a method.

Which of the following is true about constructors in OOP? * 0/1

- A) Constructors can have return types like methods.
- B) Constructors are invoked explicitly using the new keyword.
- C) Constructors are called only once during the lifetime of an object.
- D) A subclass can invoke the constructor of its superclass using the super() keyword.

Correct answer

- D) A subclass can invoke the constructor of its superclass using the super() keyword.

What will be the result of the following code * 1/1

```
int a = 5, b = 3;
if (a > b)
{
if (b > 2)
{
 System.out.println("True");
 }
}
```

- (selected) A) True ✓
- B) No output
- C) Compilation Error
- D) Runtime Exception

which of the following code snippets correctly converts a Date object to a String and vice versa using SimpleDateFormat? * 1/1

- A)Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");String formattedDate = sdf.parse(date);Date parsedDate = sdf.format(formattedDate);
- (selected) B)Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");String formattedDate = sdf.format(date);Date parsedDate = sdf.parse(formattedDate); ✓
- c)Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("MM-dd-yyyy");String formattedDate = sdf.parse(date);Date parsedDate = sdf.format(formattedDate);
- D)Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("MM-dd-yyyy");String formattedDate = sdf.parse(date);Date parsedDate = sdf.parse(formattedDate);

Consider the following code snippet. What will be the output when it is executed? class Person { String name; int age; Person(String name, int age) { this.name = name; this.age = age; } @Override public String toString() { return "Person [name=" + name + ", age=" + age + "]"; } @Override public boolean equals(Object obj) { if (this == obj) return true; if (obj == null || getClass() != obj.getClass()) return false; Person person = (Person) obj; return age == person.age && name.equals(person.name); } @Override public int hashCode() { return Objects.hash(name, age); }} public class Test { public static void main(String[] args) { Person person1 = new Person("Alice", 25); Person person2 = new Person("Alice", 25); System.out.println(person1.toString()); System.out.println(person1.equals(person2)); System.out.println(person1.hashCode() == person2.hashCode()); }} *0/1

- ○ Person [name=Alice, age=25]truetrue
- ○ Person [name=Alice, age=25]falsefalse
- ◉ Person [name=Alice, age=25]truefalse ✕
- ○ Person [name=Alice, age=25]falsetrue

Correct answer

- ◉ Person [name=Alice, age=25]truetrue

Which of the following classes in Java can be used for non-blocking I/O operations (NIO)? *0/1

- A) BufferedReader
- B) FileInputStream
- C) Selector
- D) PrintWriter

Correct answer

- C) Selector

Which of the following statements is true regarding garbage collection in Java? *0/1

- A) Java automatically runs garbage collection at fixed intervals.
- B) The garbage collector runs only when the System.gc() method is explicitly called.
- C) Objects that are unreachable or no longer referenced are eligible for garbage collection.
- D) The garbage collector cannot reclaim memory from objects that were created using new keyword.

Correct answer

- C) Objects that are unreachable or no longer referenced are eligible for garbage collection.

Which of the following statements is true about enums in Java? * 0/1

- A) An enum can inherit from another class.
- B) Enums can implement interfaces but cannot extend other classes.
- C) Enum constants are always static and final.
- D) Enums cannot have constructors or methods.

Correct answer

- B) Enums can implement interfaces but cannot extend other classes.

Which of the following is the correct way to achieve composition in Java? * 0/1

- A) A class has references to objects of other classes, but the lifetime of the referenced objects is independent of the owning class.
- B) A class has references to objects of other classes, and the lifetime of the referenced objects is controlled by the owning class.
- C) A class inherits from multiple classes.
- D) A class can have multiple objects of another class but with no strong relationship.

Correct answer

- B) A class has references to objects of other classes, and the lifetime of the referenced objects is controlled by the owning class.

What will happen if we try to access a protected variable or method outside the package but within a subclass? *0/1

- A) It will result in a compilation error, even if the subclass is in a different package.
- B) It will result in a compilation error, unless the protected member is accessed within the same package.
- C) The protected variable or method can be accessed in the subclass, even if it is in a different package.
- D) The protected member will be inherited, but access will be denied in subclasses.

Correct answer

- C) The protected variable or method can be accessed in the subclass, even if it is in a different package.

What will happen if you attempt to reassign a reference variable to a new object after it has been initialized? *0/1

- A) The original object becomes inaccessible and will be garbage collected.
- B) The reference variable will continue to refer to the original object.
- C) The new object will not be assigned to the reference variable.
- D) The original object will be deleted immediately.

Correct answer

- A) The original object becomes inaccessible and will be garbage collected.

What is the purpose of wrapper classes in Java? * 0/1

- A) To convert primitive data types to objects and vice versa.
- B) To perform arithmetic operations on primitive types.
- C) To improve memory efficiency for primitive types.
- D) To enable primitive types to be garbage collected

Correct answer

- A) To convert primitive data types to objects and vice versa.

Which of the following code snippets correctly formats a Date object into a String representation in the format dd/MM/yyyy? * 0/1

- Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("MM/dd/yyyy");String formattedDate = sdf.format(date);
- Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");String formattedDate = sdf.format(date);
- Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy");String formattedDate = sdf.format(date);
- Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("yyyy/MM/dd");String formattedDate = sdf.format(date);

Correct answer

- Date date = new Date();SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy");String formattedDate = sdf.format(date);

Which of the following methods can be used to explicitly request the JVM to run garbage collection in Java? *0/1

- (selected) A) System.gc() ✗
- B) Runtime.getRuntime().gc()
- C) System.runFinalization()
- D) All of the above

Correct answer

- D) All of the above

What will happen when you add a string literal and a new string created using the new keyword in the context of the **String pool**? *0/1

```
String str1 = "Hello";

String str2 = new String("Hello");

System.out.println(str1 == str2);
```

- (selected) A) true ✗
- B) false
- C) Compilation error
- D) Runtime exception

Correct answer

- B) false

Which method is used to assign the priority of a thread in Java? * 1/1

- a) setPriority()
- b) setThreadPriority()
- c) getPriority()
- d) assignPriority()

Which of the following statements is true about HashSet? * 0/1

- a) HashSet allows duplicate elements and maintains insertion order.
- b) HashSet does not allow null elements.
- c) HashSet is not thread-safe and does not guarantee any order of elements.
- d) HashSet maintains a sorted order of elements

Correct answer

- c) HashSet is not thread-safe and does not guarantee any order of elements.

Which of the following conditions is not required for a deadlock to occur in a system? * 0/1

- a) Mutual exclusion
- b) Hold and wait
- c) No preemption
- d) Limited number of threads

Correct answer

- d) Limited number of threads

Which of the following operators has the highest precedence in Java? * 0/1

- A) && (Logical AND)
- B) ++ (Increment)
- C) = (Assignment)
- D) == (Equality)

Correct answer

- B) ++ (Increment)

In the JVM architecture, which of the following components is responsible for loading class files into memory? *0/1

- A) Class Loader
- B) Execution Engine
- C) Garbage Collector
- D) Just-In-Time Compiler (JIT)

Correct answer

- A) Class Loader

Which of the following methods can be used to convert a String to its corresponding primitive type using a wrapper class? * 1/1

- A) parse()
- B) valueOf()
- C) toPrimitive()
- D) parseInt() for Integer, parseDouble() for Double, etc.

In Java NIO, which of the following is the correct use of the Selector class? * 0/1

- A) It is used for direct file manipulation.
- B) It allows non-blocking I/O operations on channels.
- C) It handles character encoding/decoding during input and output.
- D) It is used to map data from the disk into memory.

Correct answer

- B) It allows non-blocking I/O operations on channels.

What is the key difference between HashMap and TreeMap in Java? * 1/1

- a) HashMap maintains the order of keys, while TreeMap does not.
- b) HashMap uses a hash table for storage, while TreeMap uses a red-black tree to store keys in sorted order.
- c) HashMap allows null keys, while TreeMap does not allow null keys.
- d) HashMap can store only String keys, while TreeMap can store keys of any type.

✓ Which of the following best describes the process of serialization in Java? *1/1

- (selected) A) Converting an object into a byte stream for storage or transmission ✓
- B) Creating a shallow copy of an object
- C) Creating a deep copy of an object
- D) Transforming byte data into a human-readable format

✓ Which of the following statements about passing variables to methods is TRUE? *1/1

- A) In Java, all arguments are passed by reference, meaning any changes made to the parameter will reflect in the original variable.
- B) In Java, primitive data types are passed by reference, while objects are passed by value.
- (selected) C) In Java, primitive types are passed by value, while objects are passed by reference. ✓
- D) In Java, objects and primitive types are both passed by reference.

✓ Which of the following is the key difference between abstraction and encapsulation in Object-Oriented Programming (OOP)? *1/1

- A) Abstraction hides the internal workings of an object, while encapsulation hides the behavior.
- B) Abstraction is achieved through access modifiers, while encapsulation is achieved through methods and properties.
- (selected) C) Encapsulation hides the internal state of an object, while abstraction hides the complexity of the system. ✓
- D) There is no difference; both terms are synonymous in OOP

Which of the following interfaces must a class implement in order to support Java serialization? *1/1

- A) Cloneable
- B) Serializable
- C) Externalizable
- D) Runnable

Which of the following statements is true regarding garbage collection in Java? *1/1

- A) Java automatically runs garbage collection at fixed intervals.
- B) The garbage collector runs only when the System.gc() method is explicitly called.
- C) Objects that are unreachable or no longer referenced are eligible for garbage collection.
- D) The garbage collector cannot reclaim memory from objects that were created using new keyword.

Which of the following statements is true about the Calendar class in Java? *0/1

- A) The Calendar class is immutable and cannot be modified.
- B) Calendar is a concrete class and cannot be subclassed.
- C) Calendar is an abstract class and must be subclassed before use.
- D) Calendar objects are thread-safe without needing synchronization.

Correct answer

- C) Calendar is an abstract class and must be subclassed before use.

What is the purpose of static imports in Java? * 0/1

- A) Static imports allow you to import static variables and methods from other classes, enabling you to access them without qualifying with the class name.
- B) Static imports allow you to import static classes only.
- C) Static imports enable access to constructors from other classes.
- D) Static imports are used for importing non-static members of a class.

Correct answer

- A) Static imports allow you to import static variables and methods from other classes, enabling you to access them without qualifying with the class name.

What is the output of the following Java program? * 1/1

```
int[][] matrix = {{1, 2}, {3, 4}, {5, 6}};

System.out.println(matrix[1][1]);
```

- A) 3
- B) 4
- C) 5
- D) 6

✓ Which of the following is true about Java's primitive data types in terms of memory allocation? *1/1

- ○ A) All primitive data types in Java are allocated memory on the heap.
- ○ B) Memory allocation for primitive data types depends on the JVM's implementation and may vary across platforms.
- ◉ C) Primitive data types are stored on the stack, while objects are stored in the heap. ✓
- ○ D) All primitive data types in Java are of fixed size, and the JVM does not allow changes to their memory layout.

✓ Which of the following is true about constructor chaining in Java? * 1/1

- ○ A) Constructor chaining can only occur within the same class.
- ○ B) Constructor chaining can happen between different classes, even if they are in different packages.
- ◉ C) Constructor chaining is done using the super() keyword when invoking a constructor from a subclass. ✓
- ○ D) Constructor chaining is not possible in Java.

Which of the following statements about inheritance in Java is FALSE? * 0/1

- A) A class can inherit from multiple classes in Java (Multiple Inheritance).
- B) A subclass inherits both instance variables and methods from its superclass.
- C) A subclass can override methods of the superclass to provide its own implementation.
- D) A subclass can access the public and protected members of its superclass.

Correct answer

- A) A class can inherit from multiple classes in Java (Multiple Inheritance).

Which of the following is true about downcasting in Java? * 1/1

- A) Downcasting is always safe and does not require an explicit check.
- B) Downcasting is the act of casting a superclass reference to a subclass reference.
- C) Downcasting is allowed without any restrictions.
- D) Downcasting cannot be performed in Java.

Which of the following is NOT a valid Java token? * 0/1

- A) class
- B) @Override
- C) main
- D) ++

Correct answer

- C) main

What is the purpose of the Comparable interface in Java? * 0/1

- a) It allows you to compare objects of different classes.
- b) It enables you to define a custom sorting order for objects of a class.
- c) It provides methods to perform basic operations on collections.
- d) It allows objects to be added to a Set collection.

Correct answer

- b) It enables you to define a custom sorting order for objects of a class.

Which of the following is true about inner classes in Java? * 1/1

- (selected, correct) A) A regular inner class can access both instance and static members of the enclosing class.
- B) A method-local inner class can access only static members of the enclosing class.
- C) Anonymous inner classes do not have a name and can access instance variables directly from the enclosing class.
- D) A static inner class can access instance variables of the enclosing class without any restrictions.

NAME *

Sumit Narwade

What will be the output of the following code snippet? import java.util.*; public class Main {   public static void main(String[] args) {   List<Integer> list = new ArrayList<>(Arrays.asList(10, 20, 30, 40));   list.add(1, 25); // Insert 25 at index 1   System.out.println(list.get(1)); // What will be printed?   }} *1/1

- (selected, correct) a) 25
- b) 20
- c) 30
- d) Compilation error

Given the following code snippet, what is the result of the program? * 0/1

```
interface A {

    void methodA();

}


interface B {

    void methodB();

}


class C implements A, B {

    public void methodA() {

        System.out.println("Method A");

    }


    public void methodB() {

        System.out.println("Method B");

    }

}


public class Test {

    public static void main(String[] args) {

        C obj = new C();

        obj.methodA();

        obj.methodB();

    }
}
```

- A) Compilation Error
- B) Method A
- C) Method A followed by Method B
- D) Runtime Error

Correct answer

- C) Method A followed by Method B

What is the primary advantage of using Generics in Java? * 1/1

- a) It allows dynamic binding of types at runtime.
- b) It enables the use of primitive types in collections.
- c) It helps ensure type safety by providing compile-time type checking.
- d) It eliminates the need for reflection in Java programs.

What is the key difference between overloading and overriding in Java? * 0/1

- A) Overloading is determined at runtime, while overriding is determined at compile time.
- B) Overloading involves changing the method signature, while overriding involves changing the method body in the subclass.
- C) Overloading is not allowed in Java, whereas overriding is.
- D) Overloading occurs when the method signature is the same, and overriding happens when the return type is changed.

Correct answer

- B) Overloading involves changing the method signature, while overriding involves changing the method body in the subclass.

Which of the following is the correct way to handle multi-catch in Java? * 0/1

- (selected) A) Use multiple catch blocks for each exception type.
- B) Use a single catch block with multiple exceptions, separated by commas.
- C) Use catch (Exception | Throwable ex) to catch all types of exceptions.
- D) Use catch with | inside a try-catch block, and each exception should be an instance of Throwable

Correct answer

- B) Use a single catch block with multiple exceptions, separated by commas.

Which of the following statements about ArrayList is true? * 1/1

- a) ArrayList is synchronized by default.
- (selected) b) ArrayList allows duplicate elements.
- c) ArrayList implements Queue interface.
- d) ArrayList allows null elements, but only one at a time.

Which of the following statements is correct about the use of the final keyword in Java? * 1/1

- A) A final method can be overridden in a subclass.
- (selected) B) A final class cannot have any subclasses.
- C) A final variable can be reassigned after initialization.
- D) A final class can have a constructor.

Which of the following statements is incorrect about try-catch-finally blocks in Java? *0/1

- A) The finally block is executed even if an exception is not thrown.
- B) The finally block is executed only if the exception is caught.
- C) The finally block is executed whether or not an exception occurs, except in cases where the JVM crashes or the program exits.
- D) A finally block can be skipped if the JVM exits during the execution of the try or catch block (e.g., via System.exit()).

Correct answer

- B) The finally block is executed only if the exception is caught.

Which of the following is the correct way to create a user-defined checked exception in Java? *1/1

- A) class MyCheckedException extends RuntimeException {}
- B) class MyCheckedException extends Exception {}
- C) class MyUncheckedException extends Throwable {}
- D) class MyCheckedException extends Error {}

What happens when the finalize() method is called on an object in Java? * 0/1

- A) It is automatically called by the garbage collector before the object is reclaimed.
- B) It must be manually invoked by the developer to free up resources.
- C) It can only be called once, and it cannot be overridden in any subclass.
- D) It is called only when the object is explicitly dereferenced by the developer.

Correct answer

- A) It is automatically called by the garbage collector before the object is reclaimed.

Which of the following methods is used to pause the execution of a thread for a specified time in milliseconds? *0/1

- a) join()
- b) sleep()
- c) yield()
- d) wait()

Correct answer

- b) sleep()

In Java NIO, which of the following is used to transfer data between a FileChannel and a buffer? *0/1

- A) read() and write() methods
- B) select() and deselect() methods
- C) put() and get() methods
- D) flush() and close() methods

Correct answer

- A) read() and write() methods

What happens if an object is serialized without implementing the Serializable interface? *1/1

- A) A CloneNotSupportedException is thrown.
- B) A NotSerializableException is thrown.
- C) The object is ignored during serialization.
- D) The object is serialized as a default object.

Which of the following methods is used to remove an element from a Queue in Java? *1/1

- a) remove()
- b) pop()
- c) delete()
- d) poll()

Which of the following methods is *not* directly inherited from the Object class but is instead defined in java.util classes? *0/1

- A) clone()
- B) toString()
- C) equals()
- D) hashCode()

Correct answer

- A) clone()

Which of the following is a correct definition of a functional interface in Java? *0/1

- A) An interface that has only one abstract method and multiple default methods.
- B) An interface that has only one abstract method and no default methods.
- C) An interface that can have multiple abstract methods but only one default method.
- D) An interface that does not have any methods.

Correct answer

- A) An interface that has only one abstract method and multiple default methods.

What is the result of the following code snippet? * 0/1

```
int a = 10;

float b = 20.5f;

 a = a + b;
```

- A) Compilation Error: incompatible types
- B) a = 30.5
- C) a = 30 ✗
- D) Runtime Error

Correct answer

- A) Compilation Error: incompatible types

Which of the following statements about exceptions in Java is correct? * 0/1

- A) All exceptions inherit directly from Throwable.
- B) Error and Exception are both subclasses of Throwable, but only Error can be caught using a try-catch block.
- C) Error is a subclass of Exception, and it should be caught in the try-catch block. ✗
- D) Checked exceptions are those that are instances of Error, and unchecked exceptions are those that are instances of Exception

Correct answer

- A) All exceptions inherit directly from Throwable.

Which of the following Java 8 interface features allows defining methods with a body inside interfaces? * 1/1

- A) Abstract methods
- B) Default methods
- C) Static methods
- D) Private methods

What is the output of the following Java code, considering **inheritance** and **method overriding**? * 1/1

```
class Animal {
    void sound() {
        System.out.println("Animal sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Bark");
    }
}

public class Test {
    public static void main(String[] args) {
        Animal obj = new Dog();
        obj.sound();
    }
}
```

- A) Animal sound
- B) Bark
- C) Compilation error

D) Runtime error

Which of the following statements is true about the String pool in Java? * 0/1

A) The String pool is a separate area of memory used only for strings declared as new String().

B) Strings in the String pool are automatically garbage collected once they are no longer referenced.

C) String literals are stored in the String pool to save memory and ensure efficient string reuse.

D) The String pool is located in the heap memory, not in the method area.

Correct answer

C) String literals are stored in the String pool to save memory and ensure efficient string reuse.

What will be the output of the following **lambda expression** code? * 1/1

```
public class Test {
    public static void main(String[] args) {
        Runnable r = () -> System.out.println("Lambda Expression in Java!");
        r.run();
    }
}
```

A) Compilation error

B) Lambda Expression in Java

C) run() method not defined

D) Runnable.run() method executed

What is the behavior of the finalize() method in Java when the JVM decides to run garbage collection? *0/1

- A) The finalize() method is always called before the object is garbage collected.
- B) The finalize() method may or may not be called by the garbage collector before the object is reclaimed.
- C) The finalize() method is called when the object becomes eligible for garbage collection.
- D) The finalize() method is invoked by the developer manually before garbage collection.

Correct answer

- B) The finalize() method may or may not be called by the garbage collector before the object is reclaimed.

Which of the following statements is true about the Vector class? * 0/1

- a) Vector is not synchronized.
- b) Vector automatically doubles its size when it runs out of space.
- c) Vector has a fixed size once initialized.
- d) Vector doesn't allow dynamic resizing of elements.

Correct answer

- a) Vector is not synchronized.

Which of the following statements is TRUE about Java's memory management? *0/1

- A) Java uses manual memory management and developers are responsible for memory allocation and deallocation.
- B) Java's memory management is entirely handled by the JVM's garbage collector.
- C) The JVM allocates memory for objects but does not handle garbage collection.
- D) Java allows direct memory manipulation using unsafe methods.

Correct answer

- B) Java's memory management is entirely handled by the JVM's garbage collector.

Which of the following is NOT allowed in Java interfaces? * 0/1

- A) A class can implement multiple interfaces.
- B) An interface can extend another interface.
- C) An interface can contain concrete methods (methods with a body).
- D) An interface can contain default methods (methods with a body and a default implementation)

Correct answer

- C) An interface can contain concrete methods (methods with a body).

Which of the following interfaces is part of the Java I/O system and provides functionality for handling characters rather than raw byte data? *0/1

- A) InputStream
- B) OutputStream
- C) Reader
- D) Writer

Correct answer

- C) Reader

Which of the following is a correct way to make an object eligible for garbage collection in Java? *1/1

- A) Setting a reference variable to null
- B) Reassigning a reference variable to another object
- C) Creating an "island of isolation" by making an object inaccessible from any reachable reference
- D) All of the above

Which of the following methods is used to wait for a thread to complete execution in Java? *0/1

- a) sleep()
- b) join()
- c) wait()
- d) stop()

Correct answer

- b) join()

Consider the following code snippet: import java.util.*; public class Test { public static void main(String[] args) { List<String> list = new LinkedList<>(); list.add("a"); list.add("b"); list.add("c"); list.remove("b"); System.out.println(list); }} *1/1

What will be the output?

- a) [a, c]
- b) [b, c]
- c) [a, b, c]
- d) [a]

Which of the following statements about TreeSet is correct? * 0/1

- a) TreeSet allows duplicate elements, but sorts them in ascending order.
- b) TreeSet uses a Comparator to sort elements by default.
- c) TreeSet is based on a red-black tree, which ensures elements are stored in sorted order.
- d) TreeSet can store null elements, but only one.

Correct answer

- c) TreeSet is based on a red-black tree, which ensures elements are stored in sorted order.

Which of the following correctly describes the constructor behavior in the context of initializing reference variables in Java? *1/1

- A) Constructors are only used to initialize primitive data types.
- B) Constructors can be used to initialize reference variables, and you can use the new keyword to assign objects to them.
- C) Constructors cannot initialize reference variables, as they must always be initialized later in the program.
- D) Constructors automatically reassign all reference variables to their default values.